

CO2xComm: Porewater Sulfide

July 2026 Samples

2026-07-23

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "NO"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed

sample_month = "July"
sample_year = "2026"
Run_Date = as.Date("07/23/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner" #Instrument user
Script_run_by = "Melanie Giessner" #Code user

Run_notes="" #any notes from run

#Import Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = ("(202607|202608)(.+Plate)"))
print(Raw_plates)

## [1] "Raw Data/20260716_LTREB_H2S_Plate1.csv"
## [2] "Raw Data/20260716_LTREB_H2S_Plate2.csv"
## [3] "Raw Data/20260716_LTREB_H2S_Plate3.csv"
## [4] "Raw Data/20260716_LTREB_H2S_Plate4.csv"
## [5] "Raw Data/20260716_LTREB_H2S_Plate5.csv"
## [6] "Raw Data/20260717_LTREB_H2S_Plate10.csv"
## [7] "Raw Data/20260717_LTREB_H2S_Plate6.csv"
## [8] "Raw Data/20260717_LTREB_H2S_Plate7.csv"
## [9] "Raw Data/20260717_LTREB_H2S_Plate8.csv"
## [10] "Raw Data/20260717_LTREB_H2S_Plate9.csv"
## [11] "Raw Data/20260720_LTREB_H2S_Plate11.csv"

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- here("Porewater QAQC Logs", "GCRw_H2S_QAQC_Log.csv")

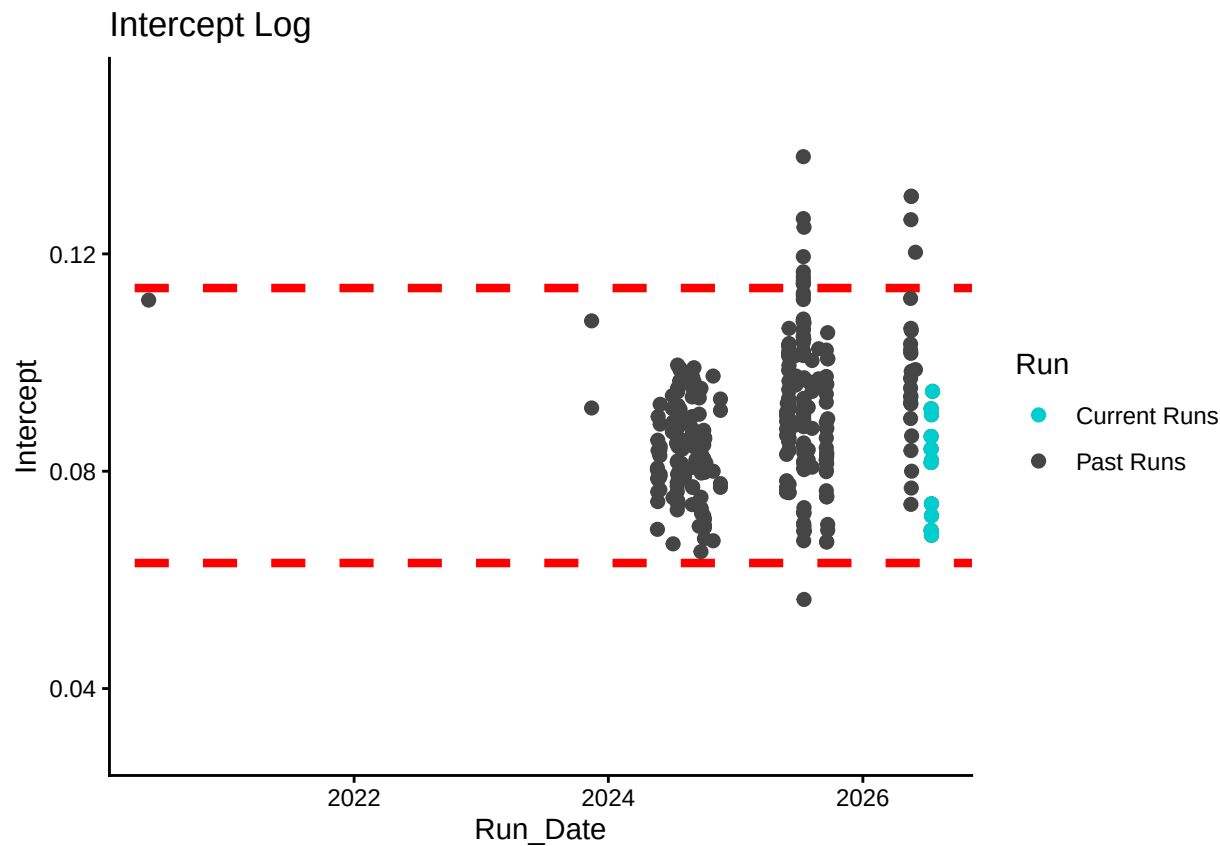
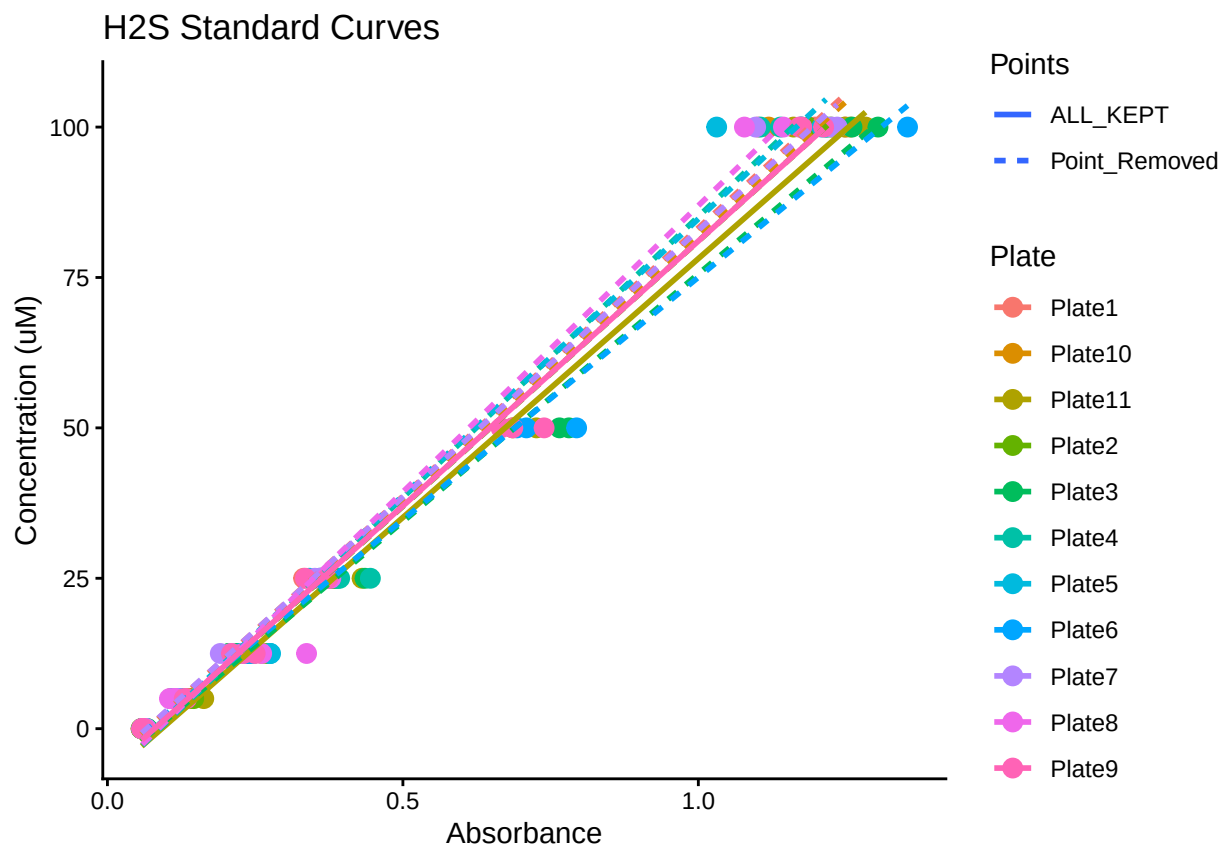
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCRw_CO2xComm_Porewater_H2S_202607.csv"
rerun_path <- "Processed Data/GCRw_CO2xComm_Porewater_H2S_202607_reruns.csv"

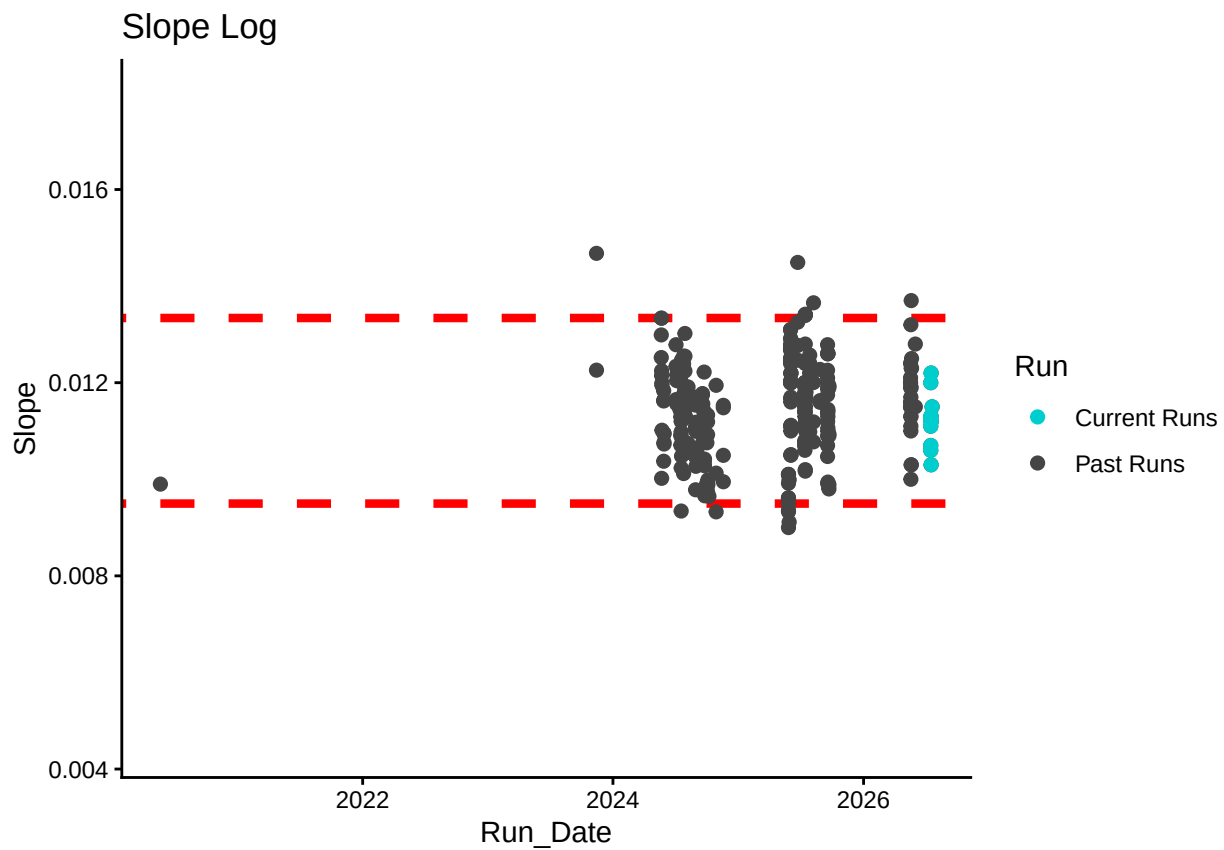
##Read in metadata and create similar sample IDs for matching to samples
```

```
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves





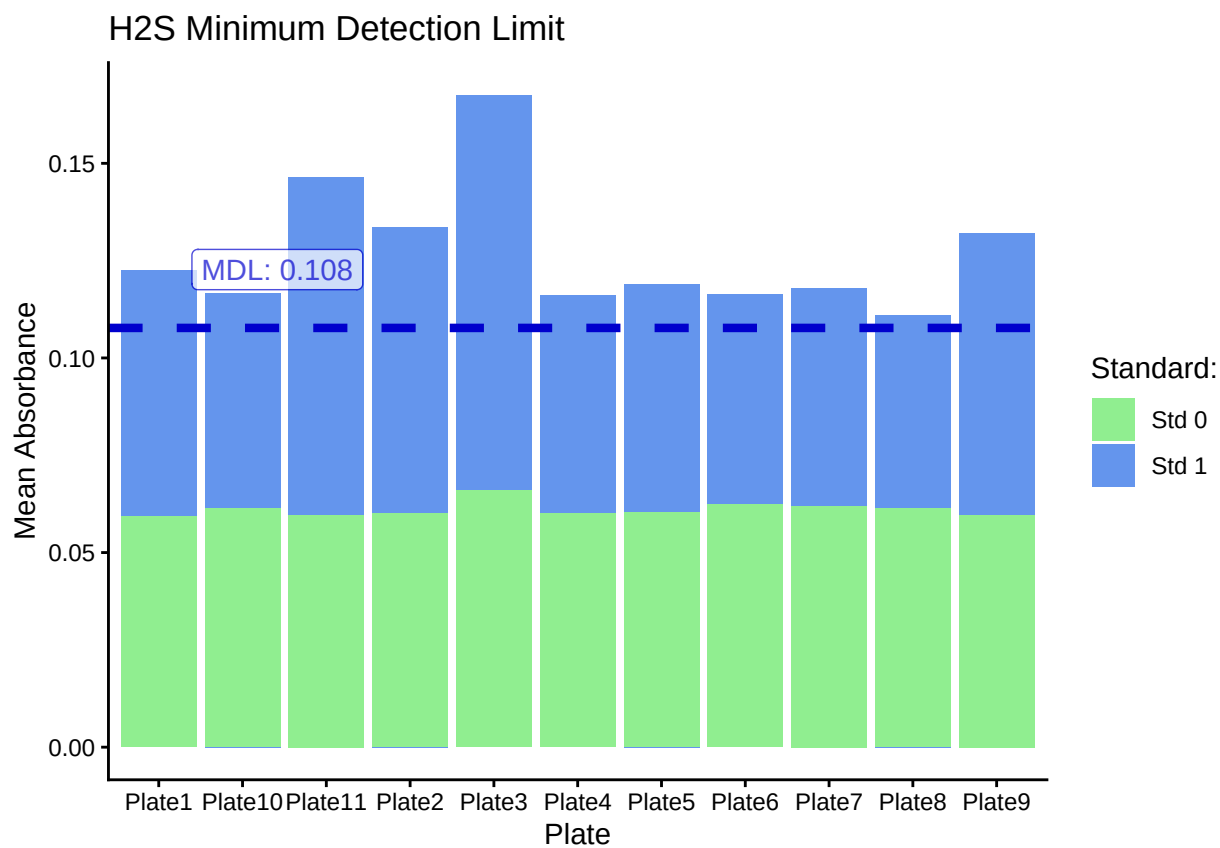
Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate10	R2_Good	CV_Good
Plate11	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Good	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Bad	CV_Good
Plate6	R2_Good	CV_Good
Plate6	R2_Good	CV_High
Plate7	R2_Good	CV_Good
Plate8	R2_Bad	CV_Good
Plate8	R2_Bad	CV_High
Plate9	R2_Good	CV_Good

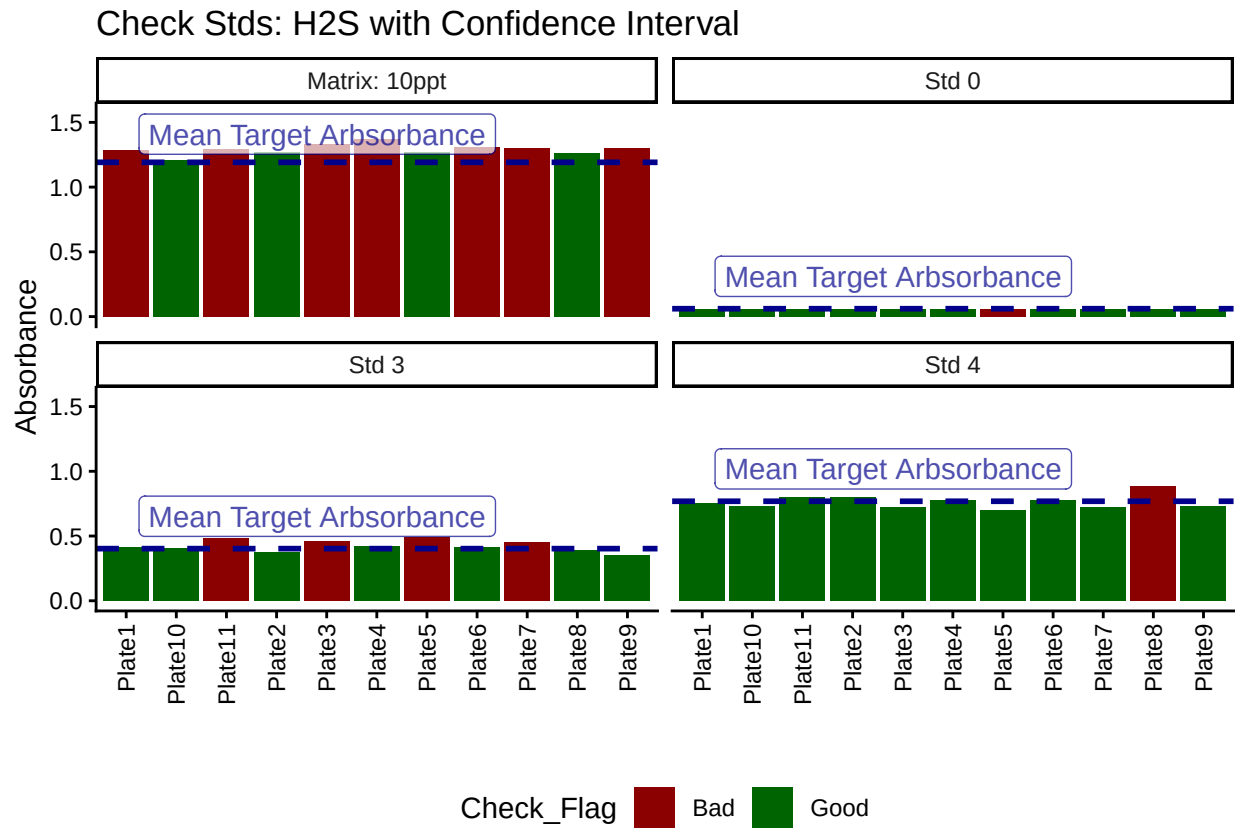
Plate	Curve_RMVD
Plate1	KEPT
Plate10	KEPT
Plate11	KEPT
Plate2	KEPT
Plate3	KEPT
Plate4	KEPT
Plate5	REMOVED
Plate6	KEPT
Plate7	KEPT

Plate	Curve_RMVD
Plate8	REMOVED
Plate9	KEPT

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

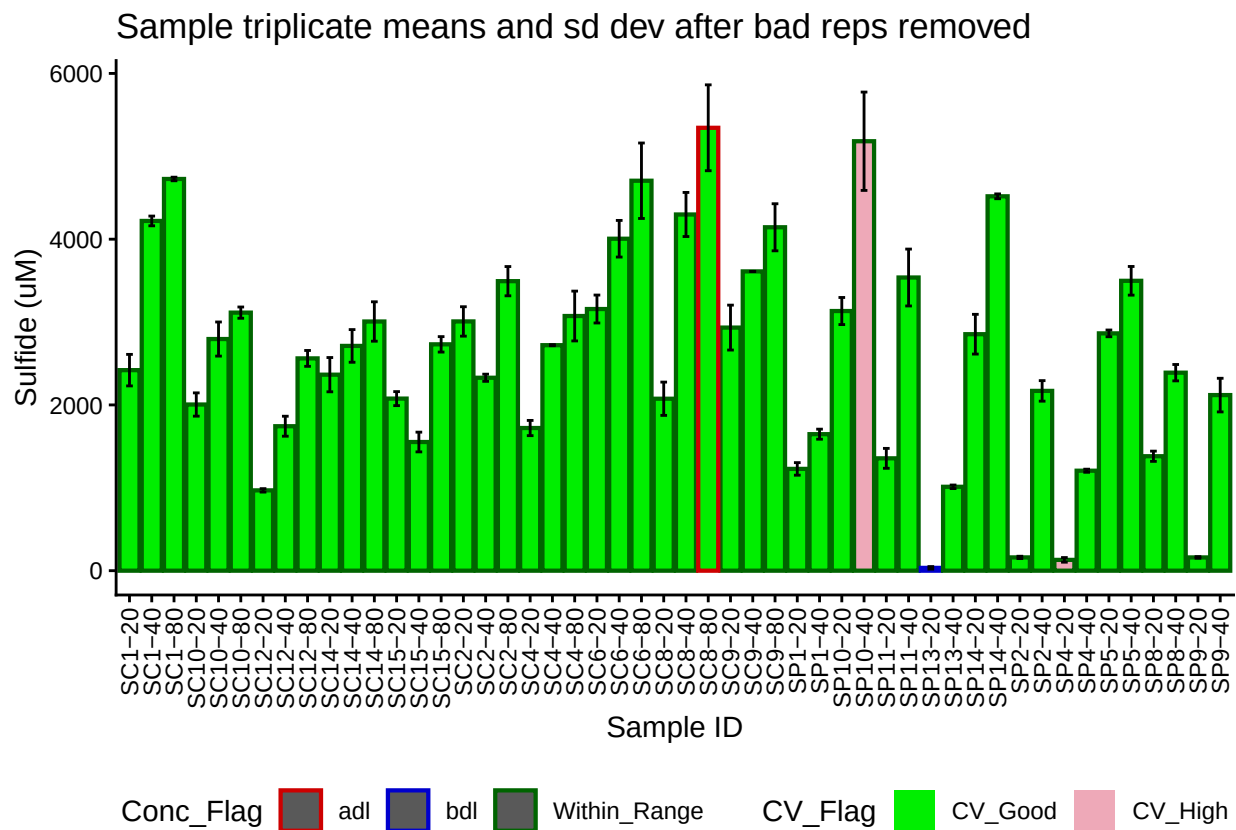
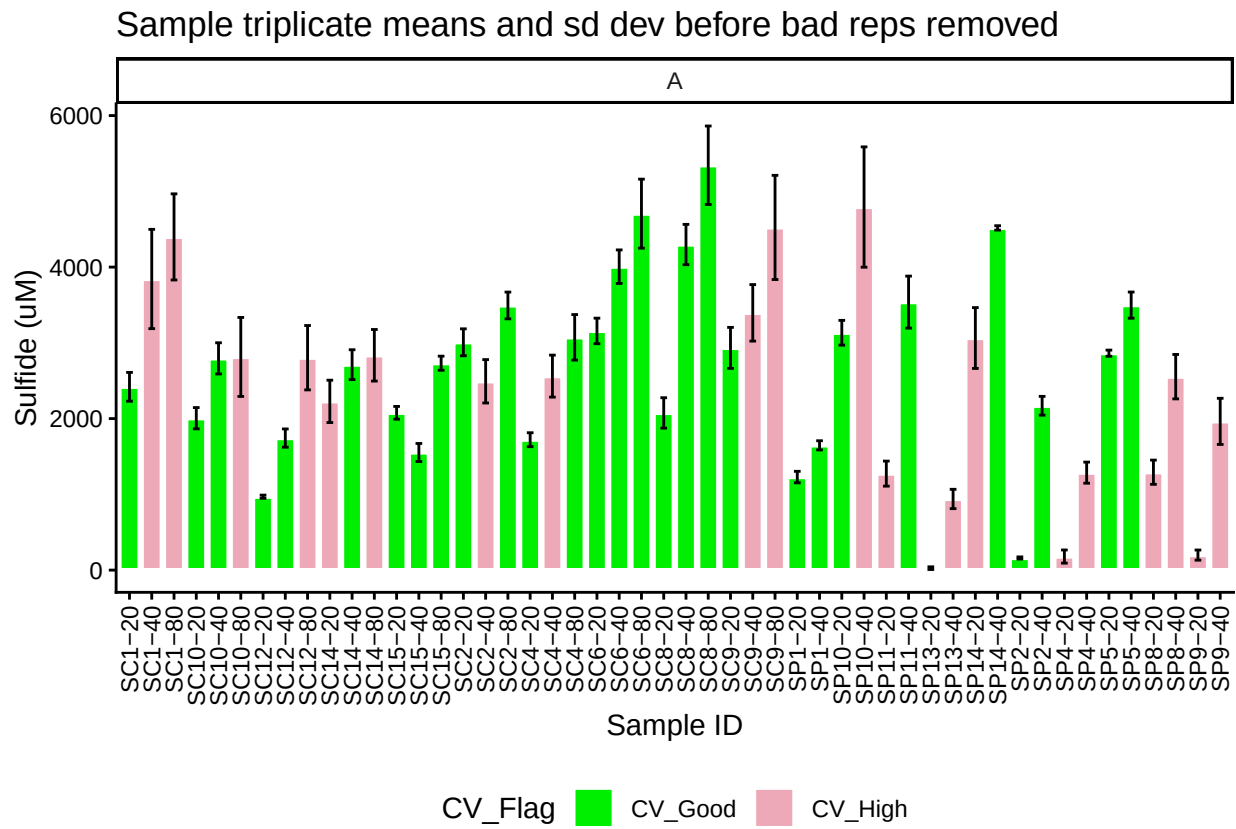
Plate	CHKs_Flag
Plate1	Std5Chk_Bad
Plate3	Std3Chk_Bad
Plate3	Std5Chk_Bad
Plate4	Std5Chk_Bad
Plate5	Std0Chk_Bad
Plate5	Std3Chk_Bad
Plate6	Std5Chk_Bad
Plate7	Std3Chk_Bad
Plate7	Std5Chk_Bad
Plate8	Std4Chk_Bad
Plate9	Std5Chk_Bad
Plate11	Std3Chk_Bad
Plate11	Std5Chk_Bad

Plate	CHK_CV_Flags
Plate1	Std3Chk_CV_High
Plate1	Std4Chk_CV_High
Plate2	Std3Chk_CV_High
Plate3	Std3Chk_CV_High
Plate4	Std4Chk_CV_High

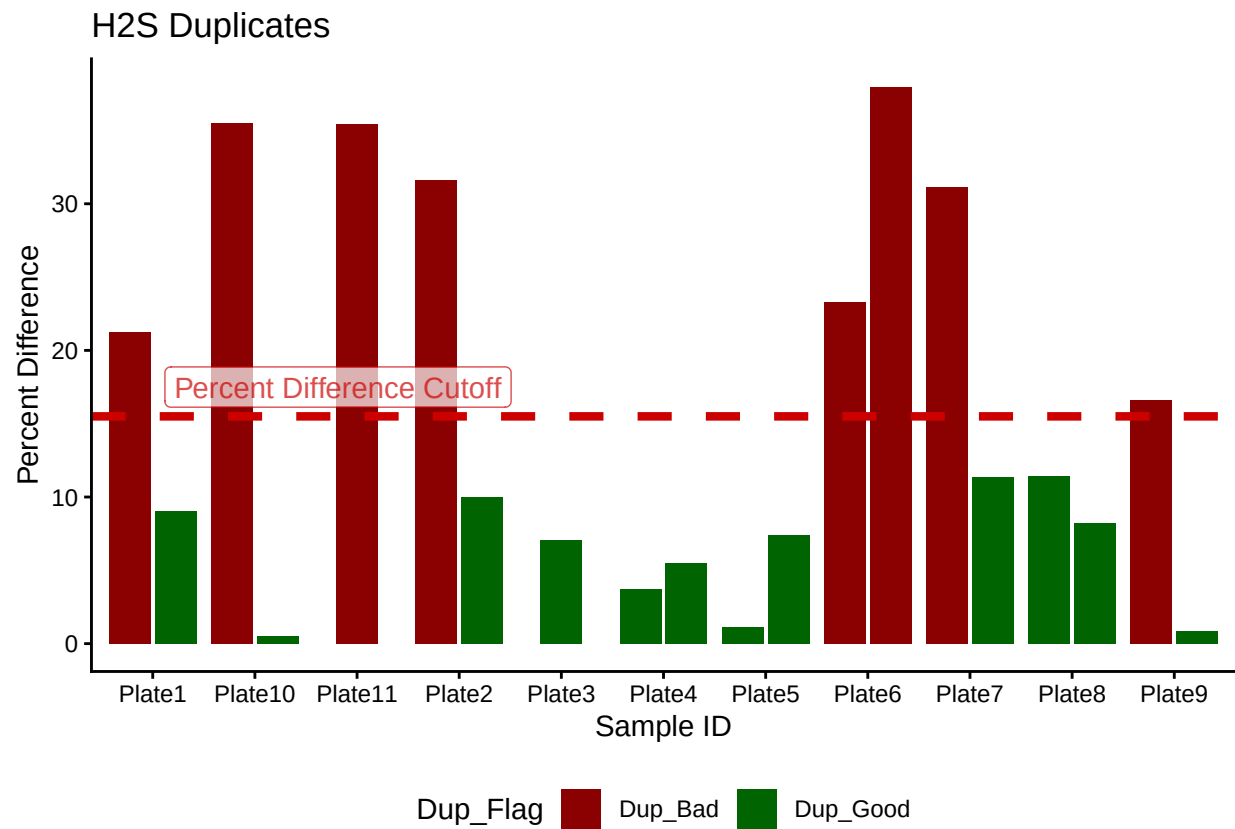
##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC



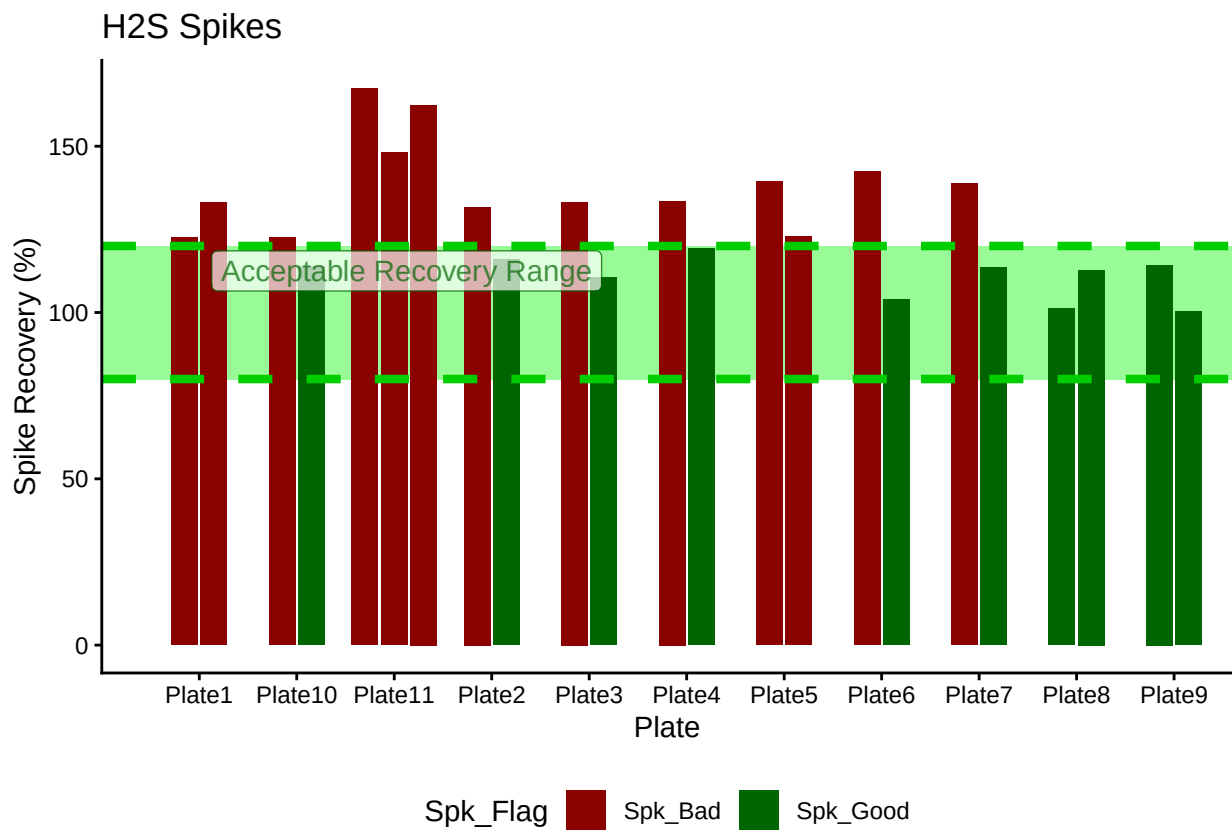
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Bad
Plate2	Dup_Bad
Plate3	Dup_Good
Plate4	Dup_Good
Plate5	Dup_Good
Plate10	Dup_Bad
Plate6	Dup_Bad
Plate7	Dup_Bad
Plate8	Dup_Good
Plate9	Dup_Bad
Plate11	Dup_Bad

Spike QAQC

```
## Warning in left_join(., spk_flags_1, by = join_by(Spk_ID, Plate)): Detected an unexpected many-to-many :
## i Row 43 of 'x' matches multiple rows in 'y'.
## i Row 2 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad
Plate3	Spk_Bad
Plate4	Spk_Bad
Plate5	Spk_Bad
Plate10	Spk_Bad
Plate6	Spk_Bad
Plate7	Spk_Bad
Plate8	Spk_Good
Plate9	Spk_Good
Plate11	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

```
## All sample IDs are present in data
```

```
##Add Flags
```

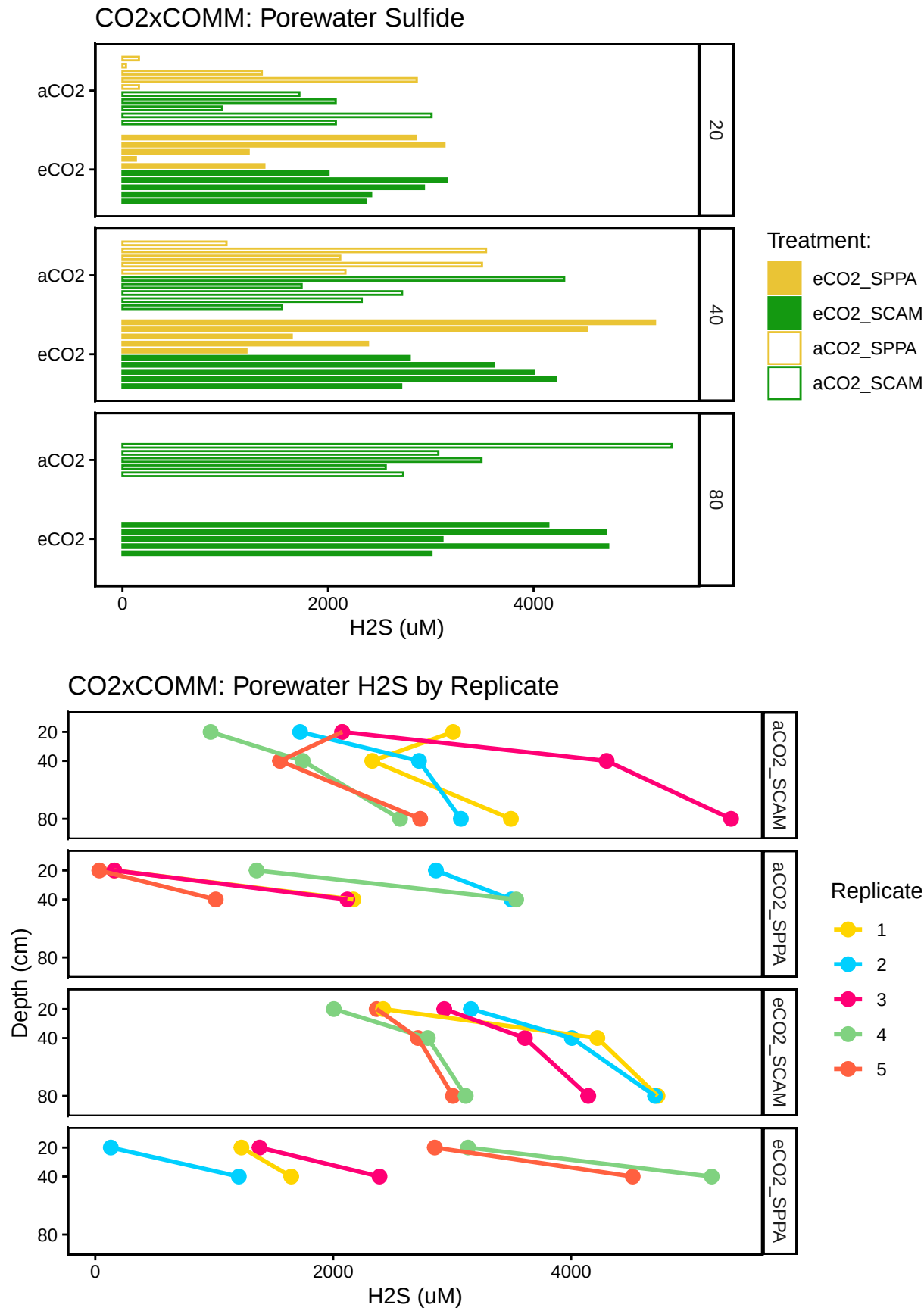
```
##Organize Data
```

```
## Adding missing grouping variables: 'Sample_ID.x'
```

```
## Adding missing grouping variables: 'Sample_ID.x'
```

Visualize Data

Visualize Data



##Export Data

Table 7: Reruns

Sample_ID.x	Conc	Conc_Flag	CV_Flag	Dilution	CV
SP4-20	130.40737	Within_Range	CV_High	50	21.299981
SC8-80	5344.03492	adl	CV_Good	50	9.697235
SP10-40	5181.73913	Within_Range	CV_High	50	11.450882
SP13-20	33.91304	bdl	CV_High	50	45.327358

END